

Multiple Choice (10 marks)

1. If N is the number of instances in the training dataset, nearest neighbors has a classification run time of
 - (a) $O(1)$
 - (b) $O(N)$
 - (c) $O(\log N)$
 - (d) $O(N^2)$
2. After applying a regularization penalty in linear regression, you find that some of the coefficients of w are zeroed out. Which of the following penalties might have been used?
 - (a) L_0 norm
 - (b) L_1 norm
 - (c) L_2 norm
 - (d) either (a) or (b)
3. As the number of training examples goes to infinity, your model trained on that data will have:
 - (a) Lower variance
 - (b) Higher variance
 - (c) Same variance
 - (d) None of the above
4. Which image data augmentation is most common for natural images?
 - (a) random crop and horizontal flip
 - (b) random crop and vertical flip
 - (c) posterization
 - (d) dithering
5. Which of the following is more appropriate to do feature selection?
 - (a) Ridge
 - (b) Lasso
 - (c) both (a) and (b)
 - (d) neither (a) nor (b)

True / False (10 marks)

Answer True or False

6. True or False: Regression does not discover causal relationships; it primarily identifies correlations between variables.
7. True or False: To calculate $P(H|E, F)$, the sufficient numbers are $P(E, F)$, $P(H)$, and $P(E, F|H)$.
8. True or False: Batch Normalization was introduced in the original ResNet paper, while Layer Normalization was not used.
9. True or False: Clustering is an effective technique for detecting fraudulent credit card transactions by grouping similar behavior patterns.
10. True or False: Overfitting is more likely when the set of training data is small, as the model may learn noise instead of general patterns.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the index of the first occurrence of a given number in a sorted array.
12. Write a function to find minimum of two numbers.

End of Paper